Adjuvants

Nonliving vaccines, especially those consisting of small molecules require the inclusion of agents to enhance their effectiveness.

These adjuvants include microbial, synthetic and endogenous preparations having adjuvant activity, but at present only aluminium or calcium salts are generally used in humans.

Adjuvants should enable antigens to be slowly released, preserve antigen integrity, target antigen presenting cells and induce cytotoxic lymphocytes.

Table 11.4: National immunization schedule

Sn. No.	Vaccine	Due age	Max age	Route
1.	BCG	At birth	till one year age	Intra dermal
2.	Hepatitis B-Birth dose	At birth	within 24 hours	Intra muscular
3.	OPV-O	At birth	within the first 15 days	Oral
4.	OPV 1, 2 & 3	At 6 weeks, 10 weeks & 14 weeks	till 5 years of age	Oral
5.	Pentavalent 1, 2 & 3 (Diphtheria + Pertuss is + Tetanus + Hepatitis B + Hib)	At 6 weeks, 10 weeks & 14 weeks	1 year of age	Intra muscular
6.	Inactivated polio vaccine	At 6 & 14 weeks	1 year of age	Intra muscular
7.	Rotavirus (where applicable)	At 6 weeks, 10 weeks & 14 weeks	1 year of age	Oral
8.	Pneumococcal conjugate vaccine (where applicable)	At 6 weeks & 14 weeks. At 9 completed months – booster	1 year of age	Intra muscular
9.	Measles/Rubella 1st dose	At 9 completed months – 12 months	5 years of age	Subcutaneous
10.	DPT Booster-1	16–24 months	7 years of age	Intra muscular
11.	Measles/Rubella 2nd dose	16–29 months	5 years of age	Subcutaneous
12.	OPV Booster	16–24 months	5 years	Oral
13.	DPT Booster – 2	5–6 years	7 years of age	Intra muscular
14.	TT	10 years & 16 years	16 years	Intra muscular

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